

Do News Outlets Link Extreme Weather to Climate Change

Group 6

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Abstract

Extreme weather events have become increasingly common, making climate change communication an important issue in news media. This project examines whether major news outlets explicitly connect extreme weather events to climate change and how such framing differs across outlets. We analyzed 221 weather-related news video transcripts from eight news organizations collected from a YouTube news dataset. After preprocessing the text, we conducted TF-IDF analysis, climate attribution scoring, topic modeling, and sentiment analysis. Our results show substantial differences in climate attribution across outlets. NBC News exhibited the highest mean climate attribution score, followed by Sky News Australia and MSNBC. Separately, MSNBC showed the highest proportion of climate-attributed articles, followed by NBC News. In contrast, CNN showed little explicit climate attribution in our sample. Topic modeling revealed themes related to extreme weather, climate discussion, disaster response, and political framing. Sentiment analysis further revealed that outlets differ in emotional tone, with Fox News showing a negative correlation between climate attribution and sentiment, suggesting a more critical framing when climate change is discussed directly. Overall, the findings suggest that news outlets frame the relationship between extreme weather and climate change in different ways, which may shape public understanding of climate-related issues.

Introduction

Extreme weather events such as hurricanes, floods, wildfires, and heatwaves have become increasingly frequent and severe in recent years. As public concern about climate change continues to grow, news media play an important role in shaping how people understand the causes and consequences of these events. For many individuals, media coverage serves as a primary source of information about climate-related issues, making the framing of extreme weather events especially influential.

Previous research has shown that news organizations often differ in how they present climate change. Some outlets frequently connect extreme weather events to broader climate trends, while others focus more on immediate impacts, disaster response, or political debates. These differences in framing may influence public perceptions of climate risks and responsibility. Understanding how climate change is discussed across media outlets can therefore provide valuable insight into the information environment surrounding environmental issues.

In this project, we examine whether major news outlets explicitly link extreme weather events to climate change and whether the strength of this connection varies across outlets. Using a dataset of 221 weather-related news video transcripts collected from eight news organizations, we apply text mining techniques including TF-IDF analysis, climate attribution scoring, topic modeling, and sentiment analysis. By comparing coverage patterns across outlets, we aim to identify differences in climate-related framing and communication.

The remainder of this paper is organized as follows. The next section reviews related literature, followed by descriptions of the dataset and methods used in the analysis. We then present the results, discuss their implications, and conclude with limitations and directions for future research.

Related Works

Previous research has demonstrated that news media play an important role in shaping public understanding of climate change. Boykoff and Boykoff (2007) examined U.S. media coverage of climate change and argued that journalistic norms can influence how scientific information is presented to the public. Their study showed that media framing may affect how audiences interpret climate-related issues.

Capstick et al. (2015) investigated public perceptions of climate change across multiple countries and found that media exposure is closely associated with climate change awareness and concern. Their findings suggest that communication channels, including news media, play a significant role in shaping public attitudes toward environmental issues.

Similarly, Carmichael and Brulle (2017) analyzed the relationship between media coverage and public concern about climate change. They found that increased media attention to climate-related topics was associated with higher levels of public concern, highlighting the influence of news coverage on public opinion.

Several studies have also examined differences in climate reporting across news organizations. Feldman et al. (2012) compared climate change coverage among CNN, Fox News, and MSNBC and found substantial variation in how climate issues were discussed and framed. Their results suggest that political orientation and editorial priorities may influence climate-related reporting.

Schäfer et al. (2014) further showed that media attention to climate change varies across political and cultural contexts. Their study demonstrated that news organizations differ not only in the amount of climate coverage they provide but also in the ways they frame climate-related topics.

While previous studies have examined climate change coverage and media framing, many focus on traditional news articles or a limited number of outlets. Our project extends this literature by analyzing weather-related video transcripts from eight major news organizations and applying text mining techniques, including TF-IDF analysis, climate attribution scoring, topic modeling, and sentiment analysis. This approach provides a quantitative comparison of how different outlets connect extreme weather events to climate change.

Data and Methods

For this project, we analyzed 221 weather-related news video transcripts from a larger YouTube news dataset provided by the course instructor. The original file included 222 videos, but one non-English transcript was removed during cleaning, leaving 221 videos for analysis. The final dataset covered eight news outlets: CNN, Fox News, MSNBC, NBC News, ABC News, CBS News, NewsNation, and Sky News Australia. The videos were originally selected with weather-related keywords, including hurricane, flood, wildfire, heatwave, and drought. The transcript from each video was used as the main source of text for analysis.

All of the analysis was done in R. We first cleaned the transcripts so that the text would be easier to compare across outlets. Then we converted the text to lowercase, separated the transcripts into individual words, removed punctuation, stop words, and common filler words, and lemmatized the remaining words. These preprocessing steps helped reduce noise and improve consistency across transcripts.

We began with Term Frequency-Inverse Document Frequency (TF-IDF) to identify distinctive vocabulary per outlet, treating each outlet as a document group to compare word use across the dataset. However, TF-IDF only captured outlet-specific language patterns and did not directly measure whether an outlet connected extreme weather to climate change, so we developed a climate attribution score for that purpose.

To measure that connection more directly, we created a climate attribution score. We used one set of climate-related keywords, such as climate change, global warming, greenhouse gas, and carbon emissions, and another set of weather-related keywords, such as hurricane, wildfire, flood, and heatwave. For each transcript, we counted how many times words from each group appeared. The attribution score was calculated by multiplying climate keyword hits by weather keyword hits. A higher score means that

climate-related language and weather-related language appeared together more often in the same transcript. In addition, we computed a climate attribution rate for each outlet, defined as the proportion of articles with a score greater than zero, which captures how frequently climate attribution appears within each outlet’s coverage and normalizes for differences in sample size across outlets.

We used Latent Dirichlet Allocation (LDA) to identify the main topics. We selected LDA over BERTopic given our small corpus, and tested K values from 3 to 8 using an 80/20 train-test split and held-out perplexity to select the optimal number of topics. Finally, we ran sentiment analysis using the sentimentr package in R, which accounts for valence shifters such as negation. Articles were classified as positive, neutral, or negative based on average sentence-level scores, and Pearson correlations were computed between attribution and sentiment scores per outlet. Since all analyses are unsupervised and descriptive rather than supervised classification, formal class balancing was not applied. Taken together, these methods allowed us to compare not only how often outlets connected climate change to extreme weather, but also how they framed those stories through word choice, topics, and tone.

Results

Climate Attribution Across News Outlets

To address our primary research question, we examined the extent to which different news outlets explicitly connected extreme weather events to climate change. Climate attribution scores were calculated based on the co-occurrence of climate-related and weather-related keywords within each transcript. Higher scores indicate stronger connections between climate change and extreme weather events.

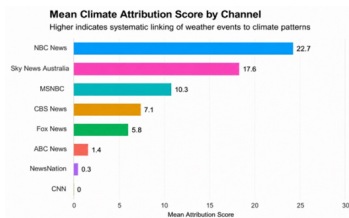


Figure 1. Mean climate attribution score by outlet. Higher scores indicate stronger connections between climate-related and weather-related language.

As shown in Figure 1, substantial differences in climate attribution were observed across news organizations. NBC News exhibited the highest mean attribution score (22.7), followed by Sky News Australia (17.6) and MSNBC (10.3). In contrast, CNN showed no explicit climate attribution within the sampled transcripts. These findings suggest that some outlets are considerably more likely than others to frame extreme weather events within a climate-change context.

Climate Attribution Rates

In addition to examining average attribution scores, we analyzed the percentage of articles that explicitly linked extreme weather events to climate change.

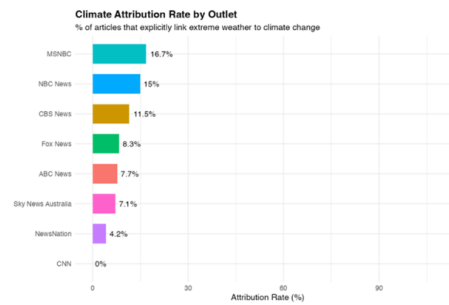


Figure 2. Percentage of climate-attributed articles by outlet. Higher percentages indicate more frequent explicit connections between extreme weather events and climate change.

As shown in Figure 2, MSNBC exhibited the highest climate attribution rate at 16.7%, followed by NBC News at 15.0% and CBS News at 11.5%. Fox News, ABC News, and Sky News Australia displayed lower rates, and none of the CNN transcripts contained explicit climate attribution. These findings further support the conclusion that climate-change framing varies substantially across news organizations.

Topic Modeling Results

To better understand the themes present in weather-related news coverage, we applied LDA topic modeling.

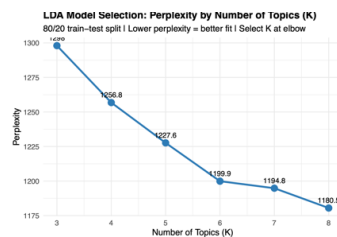


Figure 3. LDA model selection: perplexity by number of topics (K). Lower perplexity indicates better fit.

As shown in Figure 3, perplexity decreased most substantially between K=3 and K=6, after which improvement became marginal, and K=6 was therefore selected as the final model. The resulting topics reveal the major themes discussed across the dataset and provide insight into how extreme weather events are framed by news organizations.

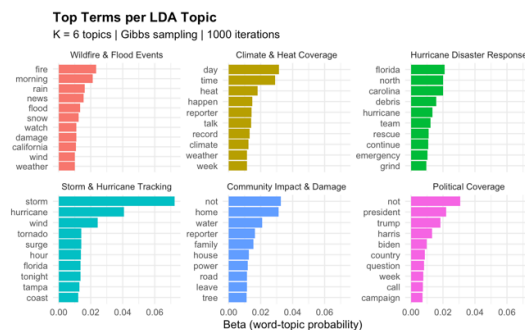


Figure 4. Top terms identified within the six LDA topics ($K = 6$).

As shown in Figure 4, the identified topics capture several recurring themes within weather-related news coverage. Wildfire and Flood Events contains terms such as fire, rain, and flood, indicating coverage of extreme weather events. Climate and Heat Coverage includes heat, climate, and record, suggesting direct climate discussion. Hurricane Disaster Response focuses on emergency management in Florida and the Carolinas, while Storm and Hurricane Tracking emphasizes severe storm events through storm, hurricane, and tornado. Community Impact and Damage reflects local reporting with home, family, and power, and Political Coverage includes political terms such as president, Trump, Harris, and Biden.

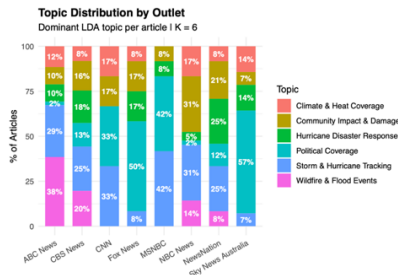


Figure 5. Topic distribution by outlet. Percentage of articles assigned to each dominant LDA topic per outlet ($K = 6$).

As shown in Figure 5, topic emphasis varied across outlets. Fox News and CNN showed higher concentrations of Political Coverage, while ABC News was dominated by Wildfire and Flood Events. Sky News Australia showed the highest proportion of Political Coverage at 57%, and Storm and Hurricane Tracking was most prominent in MSNBC.

Overall, the topic modeling results suggest that coverage of extreme weather extends beyond event reporting and frequently incorporates climate change, disaster response, and political framing. These findings indicate that climate communication occurs within multiple thematic contexts and varies substantially across news outlets.

Sentiment Analysis Results

To examine framing tone across outlets, we conducted sentiment analysis using the sentimentr package, classifying each article as positive, neutral, or negative based on average sentence-level sentiment scores.

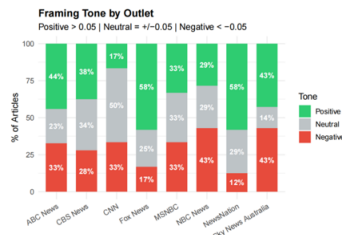


Figure 6. Distribution of positive, neutral, and negative articles across news outlets.

As shown in Figure 6, Fox News and NewsNation contained the highest proportion of positive articles at 58% each, while NBC News and Sky News Australia showed relatively more negative framing.

We also computed Pearson correlations between attribution and sentiment scores per outlet.

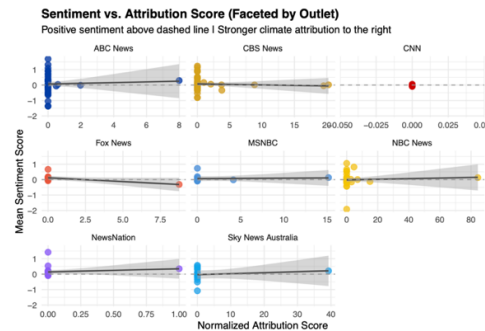


Figure 7. Sentiment versus normalized climate attribution score, faceted by outlet.

As shown in Figure 7, Fox News showed the strongest negative correlation among all outlets, suggesting its climate-related coverage tends toward more negative framing, while all other outlets showed near-zero relationships. These findings indicate that outlets differ not only in how frequently they mention climate change but also in the emotional tone of that coverage.

Discussion

The findings of this study suggest that news organizations vary substantially in how they connect extreme weather events to climate change. Consistent with our expectations, some outlets were significantly more likely to explicitly attribute extreme weather events to climate change than others. Both the climate attribution scores and attribution rates revealed clear differences across outlets, indicating that climate-related framing is not uniform within news coverage. These findings are also consistent with prior research on climate change communication and media framing (Moser, 2010).

The topic modeling results further demonstrate that weather-related reporting extends beyond the discussion of specific events. Coverage frequently incorporated themes related to climate change, disaster response, and political discourse. This finding suggests that extreme weather is often presented within broader social and environmental contexts rather than as isolated events. At the same time, the sentiment analysis revealed meaningful differences in emotional tone across outlets, indicating that organizations vary not only in how often they discuss climate change but also in how they frame those discussions. Notably, Fox News showed the strongest negative relationship between climate attribution and sentiment, suggesting that its climate-related coverage tends toward a more critical or negative tone compared to other outlets.

These findings are broadly consistent with previous research showing that media organizations differ in their approaches to climate communication and environmental reporting. The results support the idea that media framing plays an important role in shaping public understanding of climate-related issues. Because many individuals rely on news coverage as a primary source of information about climate change, differences in attribution and framing may influence how audiences perceive the causes, risks, and consequences of extreme weather events.

Limitations and Future Work

Although this study provides useful insights into climate-change framing in news media, several limitations should be acknowledged. First, the dataset consists of only 221 weather-related news transcripts from eight news organizations, which may not fully represent broader patterns of climate communication. In particular, CNN, Fox News, and MSNBC each contributed fewer than 15 transcripts, which limits the reliability of outlet-level comparisons for these organizations. Second, the analysis focuses exclusively on YouTube news transcripts, which may differ from television broadcasts, newspaper articles, or social media content.

In addition, the climate attribution score was based on keyword matching. Although this approach provides a transparent and replicable measure, it may not capture more subtle or indirect references to climate change. Similarly, topic modeling and sentiment analysis rely on automated text-analysis techniques that may not fully reflect the complexity of human interpretation. The absence of publication dates also meant that a longitudinal analysis of how climate framing shifted over time was not feasible.

Future research could expand the dataset to include additional news outlets, longer time periods, and other forms of media. Researchers may also apply more advanced natural language processing techniques, such as transformer-based language models, to better capture contextual meanings and framing patterns. Finally, future studies could examine how audiences respond to different forms of climate-related coverage in order to better understand the relationship between media framing and public perceptions of climate change.

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